COMPATIBILITY OF THE DERIVATION ACTION WITH ETALE LOCALISATION FOR THICKENINGS OF ALGEBRAIC SPACES

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 3900-3948 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 3950-3972, the complete author proof (21 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. First order thickenings of algebraic spaces, compatibility of differentials with etale localisation, and the schemes-side compatibility result; the reduction is retained.

Source reference. Original lemma lemma-first-order-thickening-maps: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 1601–1621. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-localize-differentials: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 981–996. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma more-morphisms-lemma-action-by-derivations-etale-localization: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 3972–3972. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma remark-action-by-derivations: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 3782–3788. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *Let S be a scheme. Consider a commutative diagram of first order thickenings*

$$\begin{array}{ccc} (T_2 \subset T_2') & \xrightarrow[(a_2,a_2')]{} & (X_2 \subset X_2') \\ {\scriptstyle (h,h')}\downarrow & & \downarrow{\scriptstyle (f,f')} \\ (T_1 \subset T_1') & \xrightarrow[(a_1,a_1')]{} & (X_1 \subset X_1') \end{array} \quad \text{and a commutative diagram} \quad \begin{array}{ccc} X_2' & \longrightarrow & B_2 \\ \downarrow & & \downarrow \\ X_1' & \longrightarrow & B_1 \end{array}$$

of algebraic spaces over S with $X_2 \to X_1$ and $B_2 \to B_1$ étale. For any $\mathcal{O}_{T_1}$-linear map $\theta_1 : a_1^*\Omega_{X_1/B_1} \to \mathcal{C}_{T_1/T_1'}$ let θ_2 be the composition

$$a_2^*\Omega_{X_2/B_2} = h^*a_1^*\Omega_{X_1/B_1} \xrightarrow{h^*\theta_1} h^*\mathcal{C}_{T_1/T_1'} \longrightarrow \mathcal{C}_{T_2/T_2'}$$

(equality sign is explained in the proof). Then the diagram

$$\begin{array}{ccc} T_2' & \xrightarrow[\theta_2 \cdot a_2']{} & X_2' \\ \downarrow & & \downarrow \\ T_1' & \xrightarrow{\theta_1 \cdot a_1'} & X_1' \end{array}$$

commutes where the actions $\theta_2 \cdot a_2'$ and $\theta_1 \cdot a_1'$ are as in Remark remark-action-by-derivations.

Proof. The equality sign comes from the identification $f^*\Omega_{X_1/S_1} = \Omega_{X_2/S_2}$ we get as the construction of the sheaf of differentials is compatible with étale localization (both on source and target), see Lemma lemma-localize-differentials. Namely, using this we have $a_2^*\Omega_{X_2/S_2} = a_2^*f^*\Omega_{X_1/S_1} = h^*a_1^*\Omega_{X_1/S_1}$ because $f \circ a_2 = a_1 \circ h$. Having said this, the commutativity of the diagram may be checked on étale locally. Thus we may assume T_i', X_i', B_2, and B_1 are schemes and in this case the lemma follows from More on Morphisms, Lemma more-morphisms-lemma-action-by-derivations-etale-localization. Alternative proof: using Lemma lemma-first-order-thickening-maps it suffices to show a certain diagram of sheaves of rings on X_1' is commutative; then argue exactly as in the proof of the aforementioned More on Morphisms, Lemma more-morphisms-lemma-action-by-derivations-etale-localization to see that this is indeed the case. □